第31章运行说明

返回正文 · 源码阅读

本章使用 Java 开发工具包（Java Development Kit，JDK）25.0.5.0.2、macOS AArch64和OpenJDK HotSpot。以下命令在项目根目录执行，实际工具固定为Homebrew的25版。普通编译不启动进程；所有观察为有限教学任务，没有测试、断言、压力或性能基准。共享文件和本地二进制是平台相关产物，不保证跨版本搬运。

```
P08_JDK=/opt/homebrew/opt/openjdk@25
```

示例与固定归档

CH31-E01 CacheJourney固定60,000次小请求，add和multiply是两条正常路径；无吞吐计算、断言和稳定期判据。训练与部署使用同一个Java归档（Java Archive，JAR），参数改变不重打包。

```
JAVA_HOME="$P08_JDK" /opt/homebrew/bin/mvn -pl chapters/ch31-aot-cache -am compile
"$P08_JDK/bin/jar" --create --file chapters/ch31-aot-cache/target/cache.jar --main-class
org.example.jvminternals.ch31.CacheJourney -C chapters/ch31-aot-cache/target/classes .
```

提前处理（Ahead-of-Time，AOT）三阶段

星号选项必须引用，避免zsh先作文件名展开。下面每个程序都有限结束，训练 / 装配可能比普通部署慢。记录中的time是真实单次进程口径，不能据此下统计性能结论。

```
/usr/bin/time -p "$P08_JDK/bin/java" -Xmx64m -XX:AOTMode=record -
XX:AOTConfiguration=chapters/ch31-aot-cache/target/training.aotconfig '-
Xlog:aot*=debug,jit+compilation=debug:file=chapters/ch31-aot-cache/evidence/runs/ch31-
20261103-01/training.log' -cp chapters/ch31-aot-cache/target/cache.jar
org.example.jvminternals.ch31.CacheJourney add
/usr/bin/time -p "$P08_JDK/bin/java" -Xmx64m -XX:AOTMode=create -
XX:AOTConfiguration=chapters/ch31-aot-cache/target/training.aotconfig -
XX:AOTCache=chapters/ch31-aot-cache/target/cache.aot '-Xlog:aot*=debug:file=chapters/ch31-
aot-cache/evidence/runs/ch31-20261103-01/assembly.log' -cp chapters/ch31-aot-
cache/target/cache.jar
/usr/bin/time -p "$P08_JDK/bin/java" -Xmx64m -XX:AOTMode=on -XX:AOTCache=chapters/ch31-
```

```
aot-cache/target/cache.aot '-
Xlog:aot*=debug,class+load=info,jit+compilation=debug:file=chapters/ch31-aot-
cache/evidence/runs/ch31-20261103-01/deploy.log' -cp chapters/ch31-aot-
cache/target/cache.jar org.example.jvminternals.ch31.CacheJourney add
```

25版的简化流程保留训练结束后自动启动装配子进程：

```
"$P08_JDK/bin/java" -Xmx64m -XX:AOTCacheOutput=chapters/ch31-aot-cache/target/simple.aot -
Xlog:aot=info:file=chapters/ch31-aot-cache/evidence/runs/ch31-20261103-01/simple.log -cp
chapters/ch31-aot-cache/target/cache.jar org.example.jvminternals.ch31.CacheJourney add
```

普通对照、训练外路径与画像重放

```
/usr/bin/time -p "$P08_JDK/bin/java" -Xmx64m -XX:AOTMode=off '-
Xlog:class+load=info,jit+compilation=debug:file=chapters/ch31-aot-
cache/evidence/runs/ch31-20261103-01/off.log' -cp chapters/ch31-aot-cache/target/cache.jar
org.example.jvminternals.ch31.CacheJourney add
/usr/bin/time -p "$P08_JDK/bin/java" -Xmx64m -XX:AOTMode=on -XX:AOTCache=chapters/ch31-
aot-cache/target/cache.aot '-
Xlog:aot*=debug,class+load=info,jit+compilation=debug:file=chapters/ch31-aot-
cache/evidence/runs/ch31-20261103-01/changed.log' -cp chapters/ch31-aot-
cache/target/cache.jar org.example.jvminternals.ch31.CacheJourney multiply
"$P08_JDK/bin/java" -Xmx64m -XX:AOTMode=on -XX:AOTCache=chapters/ch31-aot-
cache/target/cache.aot '-
Xlog:training=debug,aot+training=debug,jit+compilation=debug:file=chapters/ch31-aot-
cache/evidence/runs/ch31-20261103-01/profile-replay.log' -cp chapters/ch31-aot-
cache/target/cache.jar org.example.jvminternals.ch31.CacheJourney add
```

即时编译（Just-in-Time Compilation，JIT）与业务标记分开到日志，避免原先PrintCompilation与格式化输出交错。画像重放与当前编译均出现；Multiply训练外类仍从应用文件加载。缓存文件12,058,624字节，不能直接代表驻留内存或预热收益。完整日志的位置及摘要见运行记录。